



# V93000 Specifications

## Pin Scale 5000 digital card

*Published on 2021-09-29*

# Notices

---

## Legal Notices

All rights reserved. All text and figures included in this publication are the exclusive property of Advantest Corporation. Reproduction of this publication in any manner without the written permission of Advantest Corporation is prohibited. Information in this document is subject to change without notice.

## Trademarks and Registered Trademarks

- ADVANTEST is a trademark of Advantest Corporation.
- All other marks referenced herein are trademarks or registered trademarks of their respective owners.

## Warranty

The material contained in this document is provided "as is," and is subject to being changed, without notice, in future editions. Further, to the maximum extent permitted by applicable law, Advantest disclaims all warranties, either express or implied, with regard to this material and any information contained herein, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. Advantest shall not be liable for errors or for incidental or consequential damages in connection with the furnishing, use, or performance of this material or of any information contained herein. Should Advantest and the user have a separate written agreement with warranty terms covering the material in this document that conflict with these terms, the warranty terms in the separate agreement shall control.

## Technology Licenses

The hardware and/or software described in this document are furnished under a license and may be used or copied only in accordance with the terms of such license.

## Restricted Rights Legend

If software is for use in the performance of a U.S. Government prime contract or subcontract, Software is delivered and licensed as "Commercial computer software" as defined in DFAR 252.227-7014 (June 1995), or as a "commercial item" as defined in FAR 2.101(a) or as "Restricted computer software" as defined in FAR 52.227-19 (June 1987) or any equivalent agency regulation or contract clause. Use, duplication or disclosure of Software is subject to Advantest's standard commercial license terms for 93000 systems and non-DOD Departments and Agencies of the U.S. Government will receive no greater than Restricted Rights as defined in FAR 52.227-19(c)(1-2) (June 1987). U.S. Government users will receive no greater than Limited Rights as defined in FAR 52.227-14 (June 1987) or DFAR 252.227-7015 (b)(2) (November 1995), as applicable in any technical data.

## Safety Notices

**▲ CAUTION** A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

**▲ WARNING** A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

## PS5000 specifications

Use the Pin Scale 5000 digital card to execute digital tests on EXA Scale™ test systems.

### Scope of specifications

**Note**

The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).

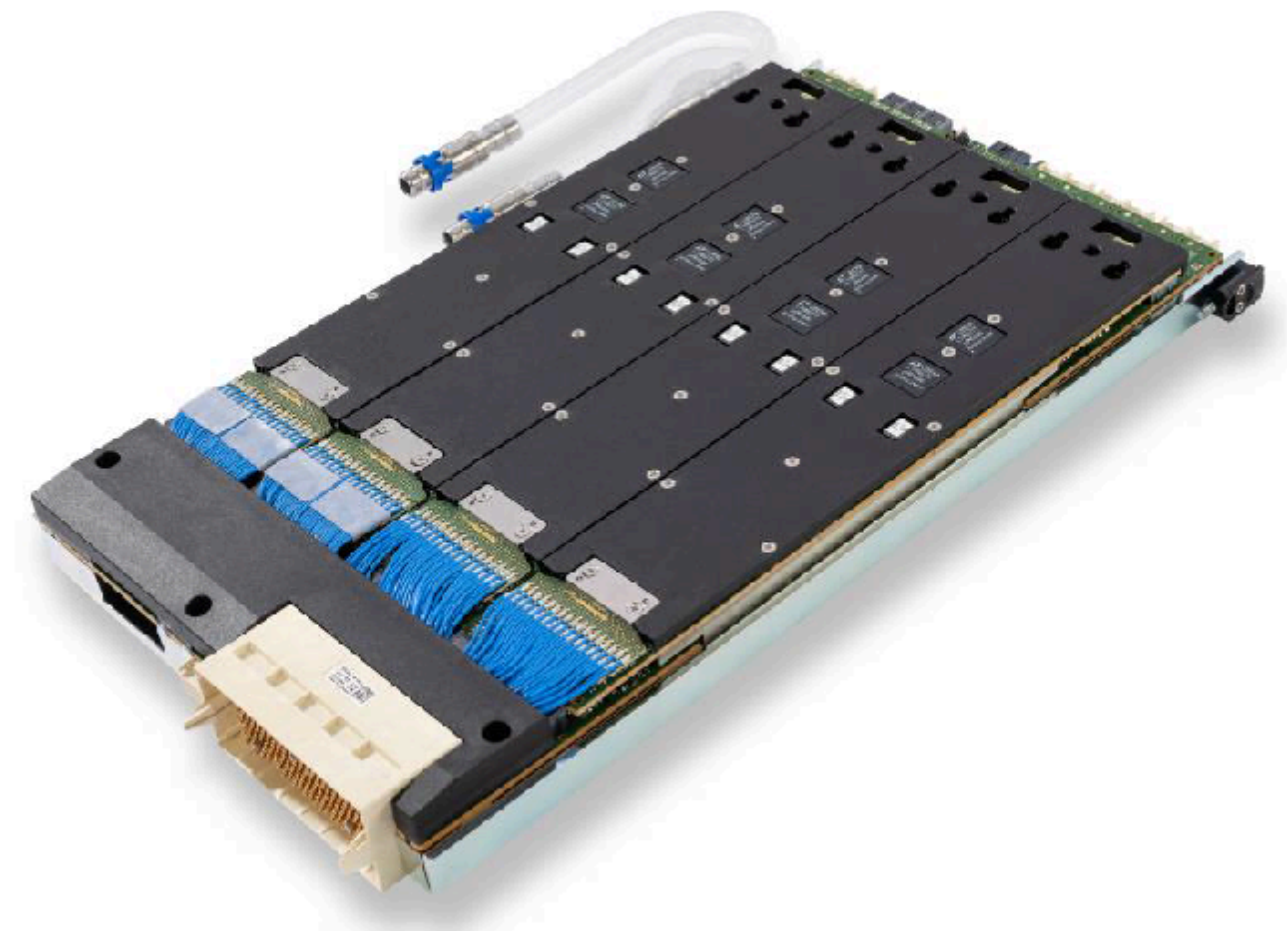
This specification covers the Pin Scale 5000, digital card (part number EK601).

Advantest specifies and verifies the specifications at the DUT interface pogo level. Appropriate DUT board design and layout including proper usage of the GND sense inputs and fixture delay measurements (TDR) is required to ensure specified performance at the device under test.

Specifications define the warranted technical performance covered by products within the stated adjustment interval and operating conditions. Typical specifications define the technical performance met by at least 80 % of all products during product qualification. A typical specification is not covered by product warranty and does not include measurement uncertainties.

### General

The Pin Scale 5000 universal digital card is an EXA Scale™ card to execute tests on digital circuitry of a DUT. It is a successor of the Pin Scale 1600 card of Smart Scale™ test systems.



*Pin Scale 5000 digital card*

***Key differences to PS1600***

- 256 instead of 128 digital channels
- Pin Scale 5000 can be equipped in EXA Scale™ test systems only
- 1,792 Mi vectors compared to 896 Mi vectors in vector storage per pin, in terms of physical memory per card this is 448 Mibyte instead of 224 Mibyte vector memory per pin
- up to 5000 Mbps instead of 1600 Mbps maximum data rate

***Further information***

For more information about the Advantest V93000 SOC Series, please visit [www.advantest.com](http://www.advantest.com).

***Contact information***

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

## Pin Scale 5000 basic parameters

### Digital channels

| Channels                                                             |                                                                                                                                   |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Maximum number of digital channels                                   | 256 per card                                                                                                                      |
| Maximum number of pseudo-random binary sequences (PRBSs) per channel | $2^{31} - 1$ , programmable seed and compare<br>User defined compliance and serial standard pattern space (64 KiByte per channel) |
| SmartLoop™                                                           | Active, re-timed, differential and single ended, loop direction changeable                                                        |

### Speed

| Speed (verification condition: half the noted voltage into 50 #) |                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum input/output data rate (pattern, PRBS, SmartLoop™)       | > 4.0 Gbps but $\leq 5.0$ Gbps at $\leq 1.0$ V (differential operation only, differential swing $\leq 1.0$ V, terminated with $100 \Omega$ , focusing functional test execution)<br>4.0 Gbps at $\leq 1.2$ V<br>2.1 Gbps at $\leq 3.0$ V<br>1 Gbps at $\leq 7.0$ V |
| Maximum clock rate                                               | 2 GHz at $\leq 1.2$ V<br>1.05 GHz at $\leq 3.0$ V<br>500 MHz at $\leq 7.0$ V                                                                                                                                                                                       |

### Operation and maintenance

For the definition of week, month, year in the context of a maintenance period refer to *Units of maintenance periods*.

| Environment, operation, and maintenance                                                                                           |                                         |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Warm-up time                                                                                                                      | 60 min                                  |
| External adjustment period                                                                                                        | 6 months                                |
| Internal adjustment period, valid at ambient temperature within $\pm 2.5$ # of internal adjustment (auto calibration) temperature | 3 months                                |
| Operating temperature range                                                                                                       | +15 # to +30 # (+59 # to +86 #)         |
| Temperature range for warranted specifications                                                                                    | +20 # to +30 # (+68 # to +86 #)         |
| Maximum humidity at +30 # (+86 #)                                                                                                 | < 70% relative humidity, non-condensing |
| Storage temperature range, without the presence of water                                                                          | -40 # to +70 # (-40 # to +158 #)        |

## Pin Scale 5000 vector memory

### General

The maximum amount of memory to be used for storing vectors is 448 MiByte per pin. The usable amount of memory depends on the given memory license. For maximum vector memory license, vector memory will be shared with a configurable amount of memory for sequencer instructions, result buffers, and level/timing sets.

Vector memory content can be shared across pins within a group of 8 pins for input, output. Result and capture memory for an output and input/output pin will still be available per individual pin. With current SmarTest software, sharing of memory content across pins is only supported for the same pin across sites.

### Functional vector modes

Different x-modes for functional vectors and scan vectors can be mixed within a pattern burst.

| Functional vector mode | Maximum data rate | Device cycles per tester cycle | Maximum timing edges per device cycle or vector | Vectors per byte | Maximum vectors per pin |
|------------------------|-------------------|--------------------------------|-------------------------------------------------|------------------|-------------------------|
| X1-mode                | 1 Gbps            | 1                              | 8 drive + 8 receive + 8 tristate                | 1                | 448 M                   |
| X2-mode                | 2 Gbps            | 2                              | 4 drive + 4 receive + 4 tristate                | 2                | 896 M                   |
| X4-mode                | 4 Gbps            | 4                              | 2 drive + 2 receive + 2 tristate                | 4                | 1,792 M                 |
| X5...X8-mode           | 5 Gbps            | 5...8                          | 1 drive + 1 receive + 1 tristate                | 5...8            | 2,240...3,584 M         |

### Scan vector modes

Every pin can be used as scan-in or scan-out pin. Switching between scan-in and scan-out within a pattern burst is supported. Scan vector mode means that due to the reduced states set, the vectors can be stored with better efficiency using either X8 (scan-in) or X5 (scan-out) mode.

| Scan vector mode | Maximum data rate | States per scan vector | Maximum scan vectors per pin | Maximum scan vectors per 8 pins | Maximum scan vectors per card |
|------------------|-------------------|------------------------|------------------------------|---------------------------------|-------------------------------|
| Scan-in          | 5 Gbps            | 0 / 1                  | 3,584 M                      | 28,672 M                        | 917,504 M                     |
| Scan-out         | 5 Gbps            | H / L / X              | 2,240 M                      | 17,920 M                        | 573,440 M                     |

## Pin Scale 5000 timing

### Accuracy

It is  $t_{EPA} = t_{OTA}/2$ . For a graphical representation refer to *Pin Scale 5000 terms* on page 22.

#### OTA and EPA

| OTA and EPA               |            |                                      |
|---------------------------|------------|--------------------------------------|
| Test system               | OTA        | ±120 ps                              |
|                           | EPA        | ±60 ps                               |
| Per card                  | OTA        | ±80 ps                               |
|                           | EPA        | ±40 ps                               |
| Per channel pair          | OTA        | ±64 ps                               |
|                           | EPA        | ±32 ps                               |
| Edge placement resolution |            | 1 ps                                 |
| Edge placement range      |            | -4 to +32 sequencer periods          |
| Clock jitter              | 2.0 GHz    | 350 fs rms (typical) <sup>[1]</sup>  |
|                           | 1.0 GHz    | 400 fs rms (typical) <sup>[1]</sup>  |
|                           | 156.25 MHz | 700 fs rms (typical) <sup>[2]</sup>  |
|                           | 100 MHz    | 1200 fs rms (typical) <sup>[2]</sup> |

### Vector period

| Vector period                                                                                  |                                                                 |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Vector period range, the change of period on the fly is supported across the full period range | 0.2 ns (5 device cycles per tester cycle, x-mode) to 100,000 ns |
| Vector period accuracy                                                                         | ±15 ppm of period setting                                       |
| Vector period resolution                                                                       | 0.001 fs                                                        |

### Eye height

Eye height is measured with the following figures:

- $V_{swing} = 500$  mV into  $50 \Omega$  at a single connection (single-ended signaling)
- $V_{swing} = 1000$  mV into  $100 \Omega$  at a pair of legs (differential signaling).

For a graphical representation refer to *Pin Scale 5000 terms* on page 22.

<sup>[1]</sup> Measurement conditions: 800 mV differential swing into 100  $\Omega$ . Fast driver is being used. Measurement: Keysight E5052B, phase noise integration range 1 kHz ... 40 MHz

<sup>[2]</sup> Measurement conditions: 800 mV differential swing into 100  $\Omega$ . Fast driver is being used. Measurement: Keysight E5052B, phase noise integration range 12 kHz ... 20 MHz

| Eye height at maximum data rates <sup>[3]</sup> |                                                             |                                                           |
|-------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| Single-ended                                    | $\Delta V_{\text{SignalEyeHeight}}$ of driver               | > 40% of $V_{\text{swing}}$ (typical)                     |
|                                                 | $\Delta V_{\text{SignalEyeHeight}}$ of driver/receiver pair | > 25% of $V_{\text{swing}}$                               |
| Differential                                    | $\Delta V_{\text{SignalEyeHeight}}$ of driver               | > 40% of $V_{\text{swing}}$ (typical) > 25% at > 4.0 Gbps |
|                                                 | $\Delta V_{\text{SignalEyeHeight}}$ of driver/receiver pair | > 25% of $V_{\text{swing}}$ > 10% at > 4.0 Gbps           |

**Transition tracking**

| Transition tracking |           |                                  |                          |
|---------------------|-----------|----------------------------------|--------------------------|
| Tracking range      | at 4 Gbps | with jitter frequency = 100 kHz  | ±128 user intervals (UI) |
|                     | at 4 Gbps | with jitter frequency = 100 MHz  | ±0.6 user intervals (UI) |
| Tracking frequency  | at 4 Gbps | with jitter amplitude = 1 UI p-p | 150 MHz                  |
|                     | at 4 Gbps | with jitter amplitude = 2 UI p-p | 37.5 MHz                 |
|                     | at 1 Gbps | with jitter amplitude = 1 UI p-p | 75 MHz                   |

**Drive jitter injection**

| Drive jitter injection |                    |
|------------------------|--------------------|
| Timing jitter range    | ±128 UI at 100 kHz |
|                        | ±0.6 UI at 100 MHz |
| Jitter frequency range | 100 kHz to 100 MHz |

<sup>[3]</sup> Single ended operation specified up to 4Gbps (see Speed on page 5). Test pattern: PRBS7, starting from clock pattern (non-static state), verification at pogo., excluding influence of performance verification board.

## Pin Scale 5000 driver

### AC performance

| AC performance (verification condition: half the noted voltage, terminated into 50 $\Omega$ ) |                          |                |        |
|-----------------------------------------------------------------------------------------------|--------------------------|----------------|--------|
| Maximum transition time (typical)                                                             | Fast mode 20% to 80%     | at 1.2 V swing | 130 ps |
|                                                                                               | Standard mode 10% to 90% | at 1.8 V swing | 450 ps |
|                                                                                               |                          | at 3.0 V swing | 750 ps |

### DC performance

| DC performance high, low, 3rd and 4th level, specified into open |                        |                                                                                                                                                               |
|------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High level (vih) range at $ V_{GS}  \leq 100$ mV                 | Fast mode              | -1.5 V to +3.5 V                                                                                                                                              |
|                                                                  | Standard mode          | -1.5 V to +7.0 V                                                                                                                                              |
| 3rd level (v3h) range at $ V_{GS}  \leq 100$ mV                  | Fast mode              | -1.5 V to +3.5 V                                                                                                                                              |
|                                                                  | Standard mode          | -1.5 V to +6.5 V                                                                                                                                              |
| Low level (vil, v3l) range at $ V_{GS}  \leq 100$ mV             | Fast mode              | -1.5 V to +3.5 V                                                                                                                                              |
|                                                                  | Standard mode          | -1.5 V to +6.5 V                                                                                                                                              |
| Level swing (vih - vil = v3h - v3l) range                        | Fast mode              | 20 mV to 1.2 V                                                                                                                                                |
|                                                                  | Standard mode          | 50 mV to 8.0 V                                                                                                                                                |
| Level resolution                                                 |                        | 1 mV                                                                                                                                                          |
| Level accuracy at $V_{GS} = 0$ V                                 | Fast mode 2-level mode | $\pm 20$ mV vih<br>$\pm 10$ mV $V_{swing} = vih - vil$<br>$\pm 20$ mV $V_{differential\ swing}$                                                               |
|                                                                  | Fast mode 4-level mode | $\pm 25$ mV vih, v3h<br>$\pm 15$ mV $V_{swing} = vih - vil = v3h - v3l$<br>$\pm 22$ mV $V_{differential\ swing}$<br>$\pm 1$ mV $\pm 1\%$ $V_{shifted\ swing}$ |
|                                                                  | Standard mode          | $\pm 5$ mV                                                                                                                                                    |

Depending on the application, v3h is also used as termination voltage vt.

The shifted swing  $V_{shifted\ swing}$  is the change of swing transitioning between vih and v3h.

### Driver impedance

| Driver impedance, including pogo cable assembly |                                                               |                            |
|-------------------------------------------------|---------------------------------------------------------------|----------------------------|
| Source impedance                                | $-1.5\text{ V} \leq vil, vih, v3l, v3h, vt \leq 3.0\text{ V}$ | $50\ \Omega \pm 3\ \Omega$ |

| Driver impedance, including pogo cable assembly |                                |                |
|-------------------------------------------------|--------------------------------|----------------|
|                                                 | 3.0 V < vil, vih, v3h, v3l, vt | 50 Ω (typical) |

## Pin Scale 5000 receiver

### AC performance

| AC performance                       |                  |
|--------------------------------------|------------------|
| Intrinsic transition time 20% to 80% | 144 ps (typical) |

### DC performance (single ended)

| DC performance (single ended)                               |         |                  |
|-------------------------------------------------------------|---------|------------------|
| Input voltage and threshold range at $ V_{GS}  \leq 100$ mV |         | -1.5 V to +6.5 V |
| Threshold voltage resolution                                |         | 1 mV             |
| Threshold voltage accuracy at $V_{GS} = 0$ mV               |         | $\pm 5$ mV       |
| Maximum input leakage current                               | High-Z  | $\pm 2$ $\mu$ A  |
|                                                             | DCL off | $\pm 10$ nA      |

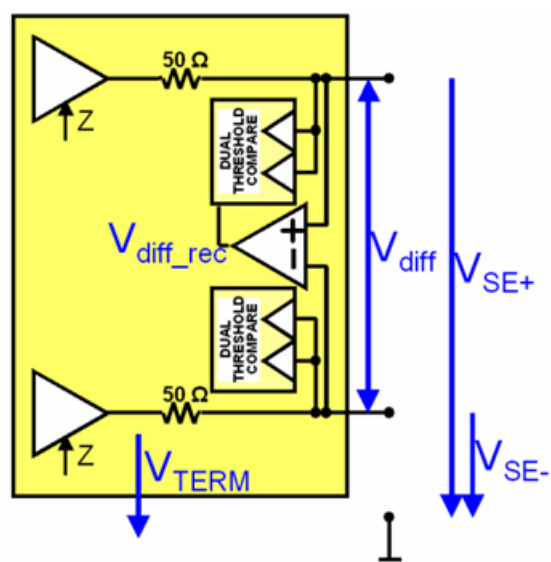
### DC performance (differential)

| DC performance (differential)                 |         |                  |
|-----------------------------------------------|---------|------------------|
| Input voltage range at $ V_{GS}  \leq 100$ mV |         | -1.5 V to +6.5 V |
| Maximum differential input voltage            |         | $\pm 8$ V        |
| Differential threshold voltage range          |         | -1.0 V to +1.0 V |
| Threshold voltage resolution                  |         | 1 mV             |
| Threshold voltage accuracy                    |         | $\pm 5$ mV       |
| Maximum input leakage current                 | High-Z  | $\pm 2$ $\mu$ A  |
|                                               | DCL off | $\pm 10$ nA      |

### Termination

| Termination                                         |                                                 |                           |
|-----------------------------------------------------|-------------------------------------------------|---------------------------|
| Single-ended impedance                              | $-1.5 \text{ V} \leq V_{SE} \leq 3.0 \text{ V}$ | $50 \Omega \pm 3 \Omega$  |
|                                                     | $V_{SE} > 3.0 \text{ V}$                        | $50 \Omega$ (typical)     |
| Center tap differential impedance                   | $-1.5 \text{ V} \leq V_{SE} \leq 3.0 \text{ V}$ | $100 \Omega \pm 6 \Omega$ |
|                                                     | $V_{SE} > 3.0 \text{ V}$                        | $100 \Omega$ (typical)    |
| Termination voltage range at $ V_{GS}  \leq 100$ mV |                                                 | -1.5 V to +6.5 V          |
| Termination voltage resolution                      |                                                 | 1 mV                      |
| Termination voltage accuracy                        |                                                 | $\pm 5$ mV                |

$V_{SE}$  is the single-ended, resulting voltage at the pin line. See the figure below.



Pin Scale 5000 voltage definitions and termination conditions

## Pin Scale 5000 time measurement unit

### DC performance

Refer to the data given in *Pin Scale 5000 receiver* on page 11.

### AC performance

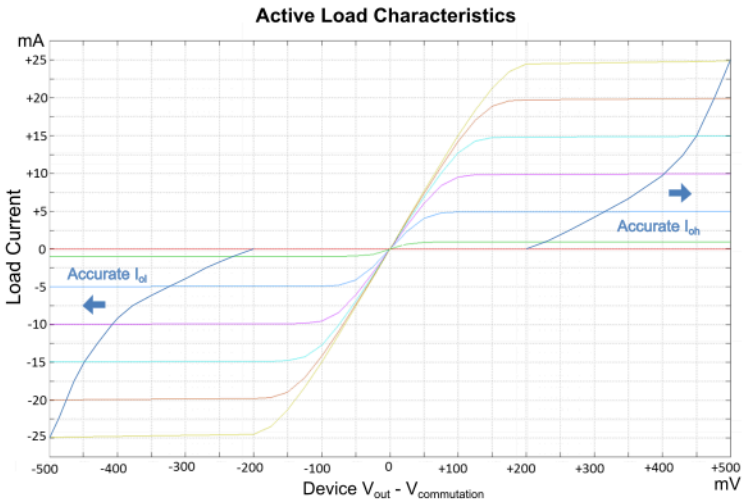
| AC performance                                      |                                  |
|-----------------------------------------------------|----------------------------------|
| Maximum input frequency                             | 2.1 GHz                          |
| Linearity error                                     | ± 2 ps                           |
| Accuracy, single timestamp to ARM (t <sub>0</sub> ) | ± 20 ps                          |
| Accuracy, skew / propagation delay (different pins) | ± 2 * (t <sub>EPA</sub> + 20 ps) |
| Intrinsic jitter                                    | 1 ps rms (typical)               |
| Resolution                                          | 1.75 ps                          |
| Maximum sampling rate                               | 375 Msps                         |

## Pin Scale 5000 programmable load

The programmable load is not available for fast mode.

| Programmable load                                                                               |                            |
|-------------------------------------------------------------------------------------------------|----------------------------|
| Load current range ( $I_{OH}$ , $I_{OL}$ )                                                      | 0 mA to 25 mA              |
| Current resolution                                                                              | 12.5 $\mu$ A               |
| Current accuracy                                                                                | $\pm 75 \mu$ A $\pm 0.5\%$ |
| Commutation voltage range ( $V_{COM}$ ) at $I_{OH}$ , $I_{OL} < 1$ mA<br>$ V_{GS}  \leq 100$ mV | -1.5 V to +6.5 V           |
| $I_{OH}$ , $I_{OL} \geq 1$ mA                                                                   | -1.0 V to +5.5 V           |
| Commutation voltage resolution                                                                  | 1 mV                       |
| Commutation voltage accuracy                                                                    | $\pm 100$ mV               |

### Active load



Pin Scale 5000 active load characteristics

## Pin Scale 5000 programmable clamp

The programmable clamp is not available for fast mode.

| Programmable clamp                                                               |                      |
|----------------------------------------------------------------------------------|----------------------|
| Clamp voltage range at $ V_{GS}  \leq 100 \text{ mV}$                            | -2.0 V to +7.0 V     |
| Maximum clamp voltage window                                                     | 9 V                  |
| Clamp voltage resolution                                                         | 1 mV                 |
| Clamp voltage accuracy, verified with a measurement current of 900 $\mu\text{A}$ | $\pm 100 \text{ mV}$ |

The sustainable total load may not exceed 5,000 mA per card.

## Pin Scale 5000 parametric measurement unit

To achieve the specified accuracy, the user must specify as minimum the number of samples given in the next sections for the averaging.

### Voltage force and measure

$I_a$  is the actual current. R is the maximum wiring resistance of 1  $\Omega$ .

| Voltage force and measure                                       |                                                                                           |                                                     |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Voltage force / measure range at $ V_{GS}  \leq 100$ mV         | $I \leq 25$ mA                                                                            | -1.50 V to +6.45 V                                  |
|                                                                 | $I > 25$ mA                                                                               | -0.60 V to +5.60 V                                  |
| Voltage force                                                   | Resolution                                                                                | 200 $\mu$ V                                         |
|                                                                 | Accuracy with force sensing at $V_{GS} = 0$ V                                             | $\pm 2$ mV                                          |
|                                                                 | Accuracy without force sensing at $V_{GS} = 0$ V                                          | $\pm (2 \text{ mV} + I_a * R)$                      |
| Voltage measure                                                 | Resolution                                                                                | 50 $\mu$ V                                          |
|                                                                 | measure (-0.2 V to +4.0 V) at $V_{GS} = 0$ V, number of internal samples for averaging 16 | $\pm (2 \text{ mV} + I_a * R)$                      |
| Accuracy ADC through PPMU                                       | measure (-1.5 V to +6.5 V) at $V_{GS} = 0$ V, number of internal samples for averaging 16 | $\pm (4 \text{ mV} + I_a * R)$                      |
|                                                                 |                                                                                           |                                                     |
| Voltage measure                                                 | Resolution                                                                                | 1 mV                                                |
|                                                                 | compare (-1.5 V to +6.5 V) at $V_{GS} = 0$ V                                              | $\pm (5 \text{ mV} + I_a * R)$                      |
| Accuracy pass/fail and SAR value through PPMU level comparators | measure (-1.5 V to +6.5 V) at $V_{GS} = 0$ V                                              | $\pm (5 \text{ mV} + I_a * R) \pm 0.5\%$ of reading |
|                                                                 |                                                                                           |                                                     |
| Capacitive loading                                              |                                                                                           | $\leq 10$ nF                                        |

### Current force and measure

| Current force and measure |                                                                                                                                   |                                                                           |                                                                             |                                                                        |                                                                       |                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Range                     | $\pm 120$ mA, ganging of both channels of a differential channel pair, this feature will be available in a later SmarTest release | $\pm 60$ mA                                                               | $\pm 1$ mA                                                                  | $\pm 100$ $\mu$ A                                                      | $\pm 10$ $\mu$ A                                                      | $\pm 2$ $\mu$ A                                                       |
| Resolution                | 40 $\mu$ A                                                                                                                        | 20 $\mu$ A                                                                | 500 nA                                                                      | 50 nA                                                                  | 5 nA                                                                  | 1 nA                                                                  |
| Measure accuracy ADC      | $\pm 100$ $\mu$ A $\pm 1\%$ , number of internal samples for averaging 4                                                          | $\pm 50$ $\mu$ A $\pm 0.5\%$ , number of internal samples for averaging 4 | $\pm 1.25$ $\mu$ A $\pm 0.5\%$ , number of internal samples for averaging 4 | $\pm 200$ nA $\pm 0.5\%$ , number of internal samples for averaging 16 | $\pm 50$ nA $\pm 0.5\%$ , number of internal samples for averaging 32 | $\pm 10$ nA $\pm 0.5\%$ , number of internal samples for averaging 64 |

| Current force and measure                       |                                 |                                 |                                |                                |                                |                               |
|-------------------------------------------------|---------------------------------|---------------------------------|--------------------------------|--------------------------------|--------------------------------|-------------------------------|
| Measure typical accuracy SAR, always one sample | $\pm 400 \mu\text{A} \pm 1.0\%$ | $\pm 200 \mu\text{A} \pm 0.5\%$ | $\pm 20 \mu\text{A} \pm 0.5\%$ | $\pm 20 \mu\text{A}$           | $\pm 2 \mu\text{A}$            | $\pm 0.5 \mu\text{A}$         |
| Force accuracy                                  | $\pm 100 \mu\text{A} \pm 0.4\%$ | $\pm 50 \mu\text{A} \pm 0.2\%$  | $\pm 5 \mu\text{A} \pm 0.5\%$  | $\pm 500 \text{ nA} \pm 0.5\%$ | $\pm 100 \text{ nA} \pm 0.5\%$ | $\pm 40 \text{ nA} \pm 0.5\%$ |

### Voltage clamp

The voltage clamp is available in current force mode only.

| Voltage clamp                                                            |                                                    |
|--------------------------------------------------------------------------|----------------------------------------------------|
| Voltage clamp range                                                      | -1.5 V to +6.5 V at $ V_{GS}  \leq 100 \text{ mV}$ |
| Clamp resolution                                                         | 1 mV                                               |
| Clamp accuracy, verified with a measurement current of 900 $\mu\text{A}$ | $\pm 100 \text{ mV}$ at $V_{GS} = 0 \text{ mV}$    |

To avoid clipping effects, the clamp must be specified with a sufficient distance to the expected measurement result. In symbolic notation,  $|\text{clamp}| > \max(|\text{expected measurement result range}|) + |\text{clamp accuracy}| + |\text{result accuracy}|$ .

### Current clamp

The current clamp is available in voltage force mode only.

| Current clamp  |                                                                                |                              |                                |                                           |                                         |                                            |
|----------------|--------------------------------------------------------------------------------|------------------------------|--------------------------------|-------------------------------------------|-----------------------------------------|--------------------------------------------|
| Current range  | $\pm 120 \text{ mA}$ , ganging of both channels of a differential channel pair | $\pm 60 \text{ mA}$          | $\pm 1 \text{ mA}$             | $\pm 100 \mu\text{A}$                     | $\pm 10 \mu\text{A}$                    | $\pm 2 \mu\text{A}$                        |
| Clamp range    | $\pm 36 \text{ mA}$ to 132 mA                                                  | $\pm 18 \text{ mA}$ to 66 mA | $\pm 0.3 \text{ mA}$ to 1.1 mA | $\pm 30 \mu\text{A}$ to 110 $\mu\text{A}$ | $\pm 3 \mu\text{A}$ to 11 $\mu\text{A}$ | $\pm 0.6 \mu\text{A}$ to 2.2 $\mu\text{A}$ |
| Resolution     | 40 $\mu\text{A}$                                                               | 20 $\mu\text{A}$             | 500 nA                         | 50 nA                                     | 50 nA                                   | 50 nA                                      |
| Clamp accuracy | $\pm 4 \text{ mA}$                                                             | $\pm 2 \text{ mA}$           | $\pm 100 \mu\text{A}$          | $\pm 10 \mu\text{A}$                      | $\pm 1 \mu\text{A}$ typical             | $\pm 0.2 \mu\text{A}$ typical              |

To avoid clipping effects, the clamp must be specified with a sufficient distance to the expected measurement result. In symbolic notation,  $|\text{clamp}| > \max(|\text{expected measurement result range}|) + |\text{clamp accuracy}| + |\text{result accuracy}|$ .

The sustainable total load may not exceed 5000 mA per card.

## Pin Scale 5000 precision voltage measure (board ADC)

### *Precision voltage measure with board ADC*

One board ADC (BADC) is shared between 16 channels for *precision* voltage measure.

| Precision voltage measure                                       |                                                                                      |                                                                                  |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Voltage range                                                   |                                                                                      | -1.5 V to +6.5 V at $ V_{GS}  \leq 100$ mV                                       |
| Measurement resolution                                          |                                                                                      | 50 $\mu$ V                                                                       |
| Measurement accuracy                                            | Ground sense connected and operational, or relative measurement without ground sense | $\pm 500$ $\mu$ V at $V_{GS} = 0$ V, number of internal samples for averaging 16 |
|                                                                 | Ground sense disconnected or not enabled                                             | $\pm 1000$ $\mu$ V, number of internal samples for averaging 16                  |
| Maximum input leakage current in voltage range -1.5 V to +6.5 V | High-Z                                                                               | $\pm 2$ $\mu$ A                                                                  |
|                                                                 | DCL off                                                                              | $\pm 10$ nA                                                                      |

The accuracy of a relative measurement refers to two consecutive measurements on the same pin or pins sharing the same board ADC, with or without ground sense.

## Pin Scale 5000 high voltage protection

### High voltage protection

| High voltage protection   |                                    |
|---------------------------|------------------------------------|
| Compliance range          | -2.5 V...+7.5 V                    |
| Overvoltage cut-off range | -80 V ... -5.5 V / +10 V ... +80 V |
| Overvoltage reaction time | ≤ 15 ns                            |
| Overcurrent cut-off range | ≥ 150 mA                           |
| Overcurrent reaction time | ≤ 500 ns                           |

## Pin Scale 5000 ground sense input

### Ground sense input

For information about  $V_{GS}$  refer to *Pin Scale 5000 terms* on page 22.

| Ground sense input         |                       |                                                                                                                          |
|----------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Ground sense inputs        | SD and HD pogo blocks | One ground sense per 16 channels                                                                                         |
|                            | VHD pogo blocks       | One ground sense per 32 channels                                                                                         |
| Ground sense voltage range |                       | $ V_{GS}  \leq 500 \text{ mV}$ , for $ V_{GS}  > 100 \text{ mV}$ voltage ranges specified have to be reduced accordingly |

## Pin Scale 5000 feature scope

---

The following features of the Pin Scale 5000 digital card are not supported by SmarTest 7.10.3 but subject to a later release:

- *time measurement unit*
- FAST mode capable of speeds up to 5.0 Gbps

**Note**

A pre-release version of FAST mode has been enabled on SmarTest 7.10.3 only for evaluation purposes.

- Using PPMU as arbitrary waveform generator (AWG)

## Pin Scale 5000 terms

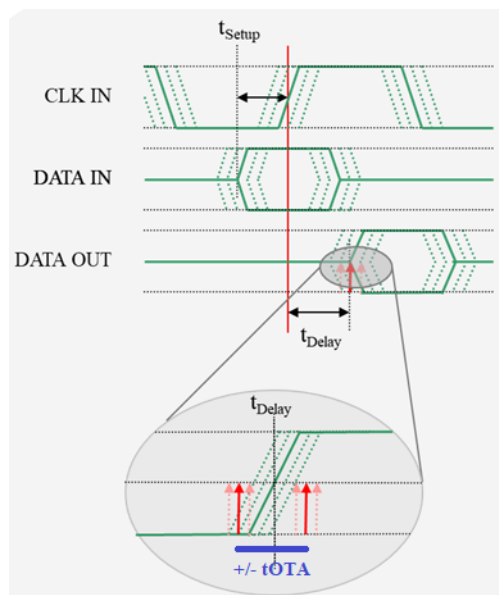
### Edge placement accuracy $t_{EPA}$

This is the worst case timing error of driver edges (or receiver edges) over all channels and settings.

It is  $t_{EPA} = t_{OTA}/2$ , refer to the explanation given below.

### Overall timing accuracy $t_{OTA}$

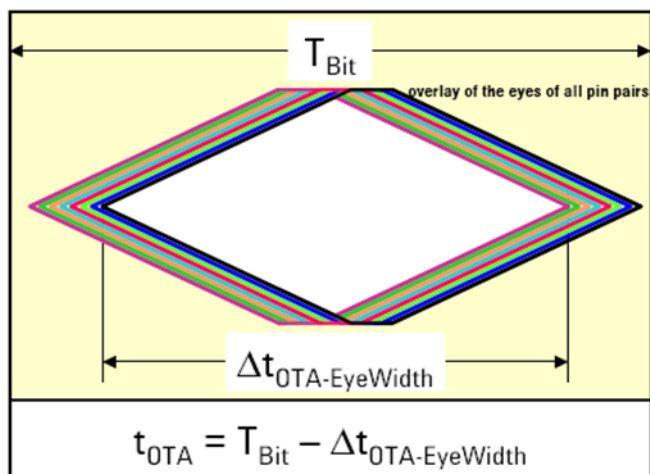
This is the worst case timing error between driver and receiver edges over all channels and settings. In the figure below,  $t_{Setup} + t_{Delay}$  is the mean timing value for which a tester edge strobe is programmed. The maximum variation of  $t_{Delay}$  is given by  $\pm t_{OTA}$ .



OTA in edge diagram

It is  $t_{OTA} = t_{EPA}$  of the driver +  $t_{EPA}$  of the receiver =  $2 * t_{EPA}$ .

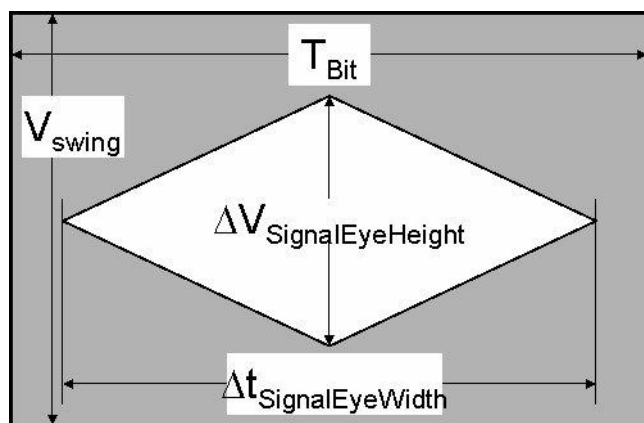
For an illustration of OTA in the eye diagram refer to the figure below.



OTA in eye diagram

### Eye height and eye width

For an illustration of the parameters  $\Delta V_{SignalEyeHeight}$  and  $\Delta t_{SignalEyeWidth}$  refer to the figure below.



Eye height and eye width

### Ground sense voltage $V_{\text{GS}}$

To support compliance to voltage force and measure specifications at the DUT pin, the PS5000 card provides *ground sense inputs* which, if properly connected to the DUT, enable to compensate for  $\geq 97\%$  (or  $\geq 94\%$  for the digital driver) of DUT ground offsets caused by test fixture GND wiring / plane resistance.

### Linearity error of AC performance

This is the maximum deviation of a periodic time stamp produced by an instrument from its *conventional true value*, given a constant conventional true value of the period between the time stamps. It is caused by non-linear characteristics of the hardware. This error refers to the timing accuracy of the I/O functions specified to be executed by the test system.

### Range

This is the interval between the minimum value and the maximum value which have been specified for a *measurand*.

### Full scale range (FSR)

The full scale range of a *measurand* is the maximum range of this measurand that can be processed according to its operational specification. If the measurand is found outside the FSR, distorting effects will have to be expected on the measurement result, such as clipping.

### Sensitivity

This is the quotient  $\Delta y / \Delta x$ , with the minimum value of a *measurand* difference  $\Delta x$  which will produce a detectable change  $\Delta y$  on the *result of the measurement*  $y$ ,  $y = f(x)$ .

### Resolution

This is the minimum value of the *measurand*  $\Delta x$  used in the definition of sensitivity.